

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

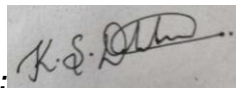
Total Marks:40

Code of Conduct:

I K.S.DHARSHANA(192412015)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Title: Accuracy Improvement of Power Transformer Faults Diagnostic Using KNN Classifier with Decision Tree Principle

Authors: Omar Kherif, Youcef Benmahamed, Madjid Teguvar, Ahmed Boubakeur and Sherif S. M. Ghoneim

Journal: IEEE Access, Volume 9, pp. 81693–81701

Year: 2021

DOI: 10.1109/ACCESS.2021.3086135

Domain/Application Area: Artificial Intelligence, Machine Learning and Power Transformer Fault Diagnosis

ML Problem: Classification of different types of faults occurring in oil-immersed power transformers using Dissolved Gas Analysis (DGA).

TASK 1 – ARTICLE SUMMARY

1(a) Article Details and Problem

The selected article focuses on diagnosing faults in oil-immersed power transformers using machine learning. The authors use Dissolved Gas Analysis (DGA), where the gases present in transformer oil are studied to identify the type of fault. The main gases considered are hydrogen (H_2), methane (CH_4), acetylene (C_2H_2), ethylene (C_2H_4) and ethane (C_2H_6). Different faults produce different combinations and concentrations of these gases.

The machine learning technique used in the paper is K-Nearest Neighbour (KNN). The authors improve the KNN approach by introducing a decision-tree principle. Instead of using one input representation for every fault, different gas-based input vectors are selected at different stages of classification.

The application is important because transformer failures can affect the reliability of electrical power systems. The paper tries to improve the accuracy of traditional DGA-based diagnosis by using an AI-based classification method.

1(b) Objective and Research Gap

The main objective of the paper is to improve the accuracy of power transformer fault diagnosis using machine learning. Traditional DGA techniques such as the Duval triangle, Rogers ratios and IEC methods are useful, but the authors point out that their diagnostic accuracy can be limited. Because of this, the paper investigates whether KNN can give better results when combined with a decision-tree-based classification strategy.

The research gap identified by the authors is that using only one type of gas representation may not be equally effective for identifying all transformer faults. Different faults have different gas characteristics, so the same input features may not work well at every stage of classification. To address this, the authors test several types of input vectors, including gas concentrations, percentage concentrations, IEC ratios, Rogers ratios, Dornenburg ratios, and Duval triangle and pentagon coordinates.

The proposed method selects the most suitable input vector for each classification stage and combines these choices with KNN. Two improved models are developed and compared. The final results show that the proposed approach can achieve an overall accuracy above 93%.

TASK 2 – METHODOLOGY ANALYSIS

2(a) ML Technique Used

The main machine learning algorithm used in the article is K-Nearest Neighbour (KNN). KNN classifies a new sample by looking at its nearest data points and assigning it to the class that has the majority representation among those neighbours.

The performance of KNN depends mainly on two things:

1. Number of neighbours, represented by k
2. Distance measure used to find neighbouring samples

The authors tested different distance measures such as:

- Euclidean
- Cityblock
- Cosine
- Correlation

They also varied the value of k from 1 to 321 during the training experiments and selected the value giving the better classification accuracy.

Decision Tree Principle

The paper does not simply use a normal decision tree as the final classifier. Instead, the decision-tree principle is used to divide the fault diagnosis into stages.

For example, electrical faults are separated first into:

PD vs. D1 & D2

and then:

D1 vs. D2

Similarly, thermal faults are divided into:

T3 vs. T1 & T2

and then:

T1 vs. T2

At each stage, the authors choose the input vector that gives better classification performance. This is the main idea behind the proposed KNN-DT method.

Dataset

The study uses a database containing 501 samples. The data were collected from previously published literature. The samples represent different transformer fault conditions.

The main input information comes from five dissolved gases:

- H_2 – Hydrogen
- CH_4 – Methane
- C_2H_2 – Acetylene
- C_2H_4 – Ethylene
- C_2H_6 – Ethane

The fault classes include:

- Partial Discharge (PD)
- Low-energy discharge (D1)
- High-energy discharge (D2)
- Thermal fault below 300°C (T1)
- Thermal fault between 300°C and 700°C (T2)
- Thermal fault above 700°C (T3)

The authors prepared different feature representations from these gases. Nine input-vector types were investigated, including raw gas concentrations, percentage concentrations, IEC ratios, Rogers ratios, Dornenburg ratios, Duval triangle coordinates, Duval pentagon coordinates and combinations of these methods.

Data Division

The 501 samples were divided into:

Purpose	Number of Samples	
Training	321	
Testing	160	
Validation	20	
Total	501	

The 20 validation samples were treated as an independent database for additional verification.

2(b) Connection with CO4

This article is strongly connected to CO4, especially the topics of Decision Trees and Statistical/Machine Learning classification mentioned in the assessment criteria.

The KNN algorithm performs classification based on similarity between data points. The decision-tree principle is then used to divide the overall fault classification into smaller classification stages.

Methodological Strength

One major strength is that the authors did not assume that one feature representation would work best for every fault. They tested several gas-based input vectors and selected suitable ones for different stages. This makes the method more flexible than simply applying KNN once to the complete dataset.

Another good point is that the researchers used an independent set of 20 samples for validation instead of reporting only training and testing performance.

Methodological Limitation

The main limitation is the relatively small dataset of 501 samples, especially when it is divided among several different fault categories. Also, the dataset was collected from existing literature rather than being collected continuously from a large number of real operating transformers. Therefore, the results may not represent every transformer condition found in actual power systems.

KNN can also become computationally expensive because it calculates distances between samples during classification. The paper itself discusses this limitation.

TASK 3 – RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The authors first tested individual input vectors with different KNN settings. For example, when the gases were used directly in ppm, the best accuracy obtained was 88.75% with correlation or cosine distance and $k = 6$.

Among the nine input vectors, the highest individual accuracy was obtained using the percentage gas concentration vector, with 91.88% accuracy using Cityblock distance and $k = 10$.

The combination of Duval triangle and pentagon coordinates achieved 90.63% accuracy. The combination of Rogers and Dornenburg ratios performed much worse, with an accuracy of 29.88%.

Comparison of Important Results

Method / Model	Accuracy
KNN using gas concentration in ppm	91.88%
KNN using combined Duval triangle + pentagon	90.63%
Proposed Model 1	92.50%
Proposed Model 2	93.75%
Model 1 – Independent validation	80%
Model 2 – Independent validation	90%

The second proposed model performed better than the first model. On the independent 20-sample validation set, Model 2 correctly diagnosed 18 out of 20 samples, giving an accuracy of 90%.

The paper also compared Model 2 with traditional DGA techniques on the same 20 validation samples:

Method	Correct Samples	Accuracy
KNN-DT Model 2	18/20	90%
Duval Triangle	6/20	30%
Rogers Ratio	4/20	25%
IEC Code	6/20	30%

This comparison shows that the proposed KNN-DT method performed considerably better on the independent validation set.

3(b) Critical Evaluation

The results are quite encouraging because the proposed method reached 93.75% accuracy during the main evaluation and 90% accuracy on the independent validation samples. The difference between these two values is important. It shows that the model did not perform exactly the same on unseen validation data, but it still produced a reasonably strong result.

I think the independent validation is one of the stronger parts of the study because it gives a better idea of how the method performs on data that were not used in developing the main models.

However, there are some limitations in interpreting the accuracy. The complete dataset contains only 501 samples, and only 20 samples were used for the independent validation. A validation set of 20 samples is quite small, so even one or two incorrect predictions can change the reported percentage significantly.

Another issue is that the data were collected from previous studies. The characteristics of transformers can vary depending on their age, design, operating conditions and environment. Therefore, testing the method on a much larger dataset collected from real transformers in different locations would make the conclusion stronger.

The authors could also strengthen the work by using k-fold cross-validation, reporting a confusion matrix, precision, recall and F1-score for every fault class, and testing the model on datasets from different utilities or transformer manufacturers. These experiments would give a clearer picture of whether the model is consistently good or mainly performs well on this particular dataset.

3(c) Real-World Applicability

The proposed technique can be useful in power-grid monitoring and predictive maintenance. Transformer oil samples can be periodically analysed using DGA, and the resulting gas concentrations can be given to the machine learning model. The system could then help engineers identify the likely type of transformer fault at an early stage.

This could help reduce unexpected transformer failures and improve maintenance planning. Early fault detection is especially useful because transformer failure can lead to expensive repairs and interruptions in power supply. The article itself explains that DGA is already a widely used technique for identifying abnormal transformer conditions.

However, applying the system on an industrial scale would not be completely simple. A large and reliable database of real transformer data would be needed. Different transformers may also behave differently, so a model trained on one dataset may not work equally well everywhere.

There are also practical issues such as the cost of regular DGA testing, data quality, sensor accuracy and the need for engineers to verify important decisions. For a critical electrical system, the ML model should support human experts rather than being the only basis for maintenance decisions.

TASK 4 – REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights I Gained

1. Feature selection can strongly affect classification

One thing I understood better from this article is that choosing the input features is very important in machine learning. The authors tried nine different representations of the same DGA information and found large differences in accuracy. For example, one representation gave about 92% accuracy while another gave only about 30%.

This connects with the Statistical Learning and Classification concepts in CO4 because the quality and representation of input data can directly affect the performance of a classifier.

2. KNN depends on the choice of neighbours and distance

Before reading this paper, I mainly understood KNN as a method that classifies a sample based on its nearest neighbours. This article made it clearer that the choice of k and distance measure can significantly change the result.

The authors tested different distance measures and different values of k and selected the combinations that produced better accuracy. This connects directly with the machine learning classification techniques covered in CO4.

3. Decision-tree thinking can simplify a classification problem

Another important point I learned is that a complicated multiclass problem can be divided into smaller decisions. Instead of asking the model to identify all transformer faults at once, the authors first separate electrical and thermal faults and then classify them further.

This connects with the Decision Tree topic in CO4. It helped me understand how hierarchical classification can make a difficult classification problem easier to handle.

4(b) Proposed Extension

If I were extending this research, I would test the KNN-DT method using a larger real-world transformer dataset collected from different power stations.

The existing study uses 501 samples and only 20 independent validation samples. A larger dataset would contain more examples of different transformer conditions and would make the evaluation more reliable.

I would also compare the proposed KNN-DT model with newer classification techniques such as Random Forest, SVM and Gradient Boosting using the same dataset. The models could be evaluated using accuracy, precision, recall, F1-score and confusion matrices instead of depending mainly on accuracy.

I think this experiment is feasible because the existing DGA measurements can still be used as the input data. The main improvement would be increasing the amount and diversity of the data. If the proposed method continues to perform well on data from different transformers, it would have stronger potential for real-world predictive maintenance.

OVERALL REVIEW

Overall, I found this article useful because it shows a practical application of machine learning rather than discussing classification only in theory. The most interesting part for me was the way the authors combined KNN with a decision-tree principle. Instead of treating every fault in exactly the same way, they selected different input representations for different stages of the diagnosis.

The reported results are promising, especially the 93.75% accuracy of Model 2 and its 90% accuracy on the independent validation set. At the same time, the small dataset and small independent validation set mean that the results should not be treated as final proof that the method will work equally well for every transformer.

Overall, the paper gave me a better understanding of how feature representation, KNN parameters and decision-tree-based classification can work together to solve a real-world classification problem. It also showed me that a high accuracy value alone is not enough; the dataset and evaluation method also need to be considered before deciding whether a machine learning model is reliable.

Reference

Kherif, O., Benmahamed, Y., Teguier, M., Boubakeur, A., & Ghoneim, S. S. M. (2021). *Accuracy Improvement of Power Transformer Faults Diagnostic Using KNN Classifier with Decision Tree Principle*. IEEE Access, 9, 81693–81701. DOI: 10.1109/ACCESS.2021.3086135.